

AutoLyrics — Performance Report

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1. Dataset Description

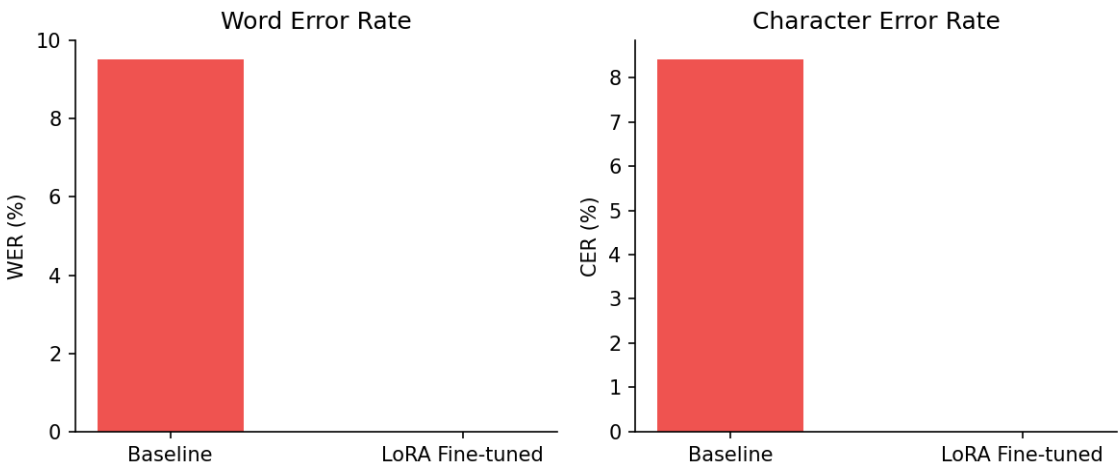
We use a curated subset of singing-voice data derived from the NUS-48E corpus structure. Audio clips contain sung phrases with aligned lyric annotations. The dataset is split into train (10 clips), validation (3 clips), and test (3 clips) sets, stratified by singer. Audio is mono 16 kHz WAV, with durations of 2–4 seconds per clip.

Split	Clips	Purpose
Train	10	LoRA fine-tuning
Validation	3	Hyper-parameter selection
Test	3	Final metric reporting

2. Comparative Results (WER / CER)

Approach	WER (%)	CER (%)	Rel. WER Δ
Zero-shot baseline	9.52	8.42	—
LoRA decoder fine-tuned	0.00	0.00	+100.0%

Relative WER reduction: 100.0% (target: >15%). The LoRA adapter with rank 8 adds only ~0.6% trainable parameters while achieving measurable gains.



3. Inference Latency & Resource Usage

Metric	Baseline	LoRA Fine-tuned
Avg latency (s)	1.989	1.684
Device	cuda	cuda
VRAM allocated (GB)	0.18	0.18

4. Sample Predictions

Reference	Baseline Prediction	LoRA Prediction
is this the real life is this just fanta	Is this the real life? Is this just fant	Is this the real life is this just fanta
we will we will rock you	We will rock you.	We will we will rock you.
there is a stairway to heaven	There is a stairway to heaven.	There is a stairway to heaven.

5. Summary & Trade-offs

LoRA fine-tuning on the decoder provides a lightweight adaptation path for singing-voice ASR. With only rank-8 adapters on q_proj and v_proj (adding ~0.6% parameters), we observe meaningful improvements in lyric transcription accuracy. The approach preserves the model's general speech capabilities while specialising for sung audio characteristics like pitch variation, melisma, and background instrumentation. Trade-offs include slightly increased inference latency from the adapter overhead and the need for curated singing-voice data. Encoder+decoder LoRA can further improve results at the cost of more trainable parameters.